

[BINNING]

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NOTE: ALWAYS SORT THE DATA

Q. DATA = [4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34]

NUMBER OF BINS = 3 (m)

NUMBER OF ITEMS = 12 (n)

= PARTITION INTO EQUAL FREQUENCY:

$$n = 12$$

$$m = 3$$

$$\text{EQUAL FREQUENCY} = n/m = 12/3 = 4$$

$$\text{BIN 1} = [4, 8, 9, 15]$$

$$\text{BIN 2} = [21, 21, 24, 25]$$

$$\text{BIN 3} = [26, 28, 29, 34]$$

= SMOOTHING BY BIN MEANS:

$$\text{BIN 1} \Rightarrow m_1 = [4 + 8 + 9 + 15] / 4$$

$$= 21.75 \approx 22$$

$$\text{BIN 2} \Rightarrow m_2 = [21 + 21 + 24 + 25] / 4$$

$$= 22.75 \approx 23$$

$$\text{BIN 3} \Rightarrow m_3 = [26 + 28 + 29 + 34] / 4$$

$$= 29.25 \approx 29$$

$$\text{BIN 1} = [22, 22, 22, 22]$$

$$\text{BIN 2} = [23, 23, 23, 23]$$

$$\text{BIN 3} = [29, 29, 29, 29]$$

= SMOOTHING BY BIN MEDIAN:

$$\text{BIN 1} \Rightarrow k_1 = [8 + 9] / 2 = 8.5$$

$$\text{BIN 2} \Rightarrow k_2 = [21 + 24] / 2 = 22.5$$

$$\text{BIN 3} \Rightarrow k_3 = [28 + 29] / 2 = 28.5$$

$$\text{BIN 1} = [8.5, 8.5, 8.5, 8.5]$$

$$\text{BIN 2} = [22.5, 22.5, 22.5, 22.5]$$

$$\text{BIN 3} = [28.5, 28.5, 28.5, 28.5]$$

= SMOOTHING BY BIN BOUNDARIES:

$$\text{BIN 1} = [4, 4, 4, 15]$$

$$\text{BIN 2} = [21, 21, 25, 25]$$

$$\text{BIN 3} = [26, 26, 26, 34]$$

[K-MEANS - 1 DIMENSION]

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NOTE: ALWAYS SORT THE DATA

Q. DATA = [2, 3, 6, 8, 9, 12, 15, 18, 22]

NUMBER OF ITEMS (n) = 9

NUMBER OF CLUSTERS (m) = 3

LET,

$$C1 = 2$$

$$C2 = 9$$

$$C3 = 22$$

①	CLUSTER	OBJECT	CENTROID
	C1 (2)	2, 3	2.5
	C2 (9)	6, 8, 9, 12, 15	10
	C3 (22)	18, 22	20

AVG

$$C1 \Rightarrow (2+3)/2 = 2.5$$

$$C2 \Rightarrow (6+8+9+12+15)/5 = 10$$

$$C3 \Rightarrow (18+22)/2 = 20$$

②	CLUSTER	OBJECT	CENTROID
	C1 (2.5)	2, 3, 6	3.667
	C2 (10)	8, 9, 12, 15	11
	C3 (20)	18, 22	20

AVG

$$C1 \Rightarrow (2+3+6)/3 = 3.667$$

$$C2 \Rightarrow (8+9+12+15)/4 = 11$$

$$C3 \Rightarrow (18+22)/2 = 20$$

③	CLUSTER	OBJECT	CENTROID
	C1 (3.667)	2, 3, 6	3.667
	C2 (11)	8, 9, 12, 15	11
	C3 (20)	18, 22	20

CLUSTERS = [[2, 3, 6], [8, 9, 12, 15], [18, 22]]

[FIVE NUMBER SUMMARY]

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NOTE: ALWAYS SORT THE DATA

Q. DATA = [5, 10, 13, 15, 16, 16, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70, 85]

$$n = 28$$

a) MEAN = $\sum x / n = 890 / 29 = 30.68$

b) MEDIAN = $(n+1) / 2 = 30 / 2 = 15^{\text{th}} \text{ VALUE} = 25$

c) FIRST QUARTILE (Q1) = MEDIAN OF FIRST HALF TILL MEDIAN = $(20+20) / 2 = 20$

d) THIRD QUARTILE (Q3) = MEDIAN OF SECOND HALF FROM MEDIAN = $(35+36) / 2 = 35.5$

e) STANDARD DEVIATION = $\sqrt{\frac{\sum x_i - \bar{x}}{n-1}} = 17.186$

f) BOX AND WHISKER PLOT:

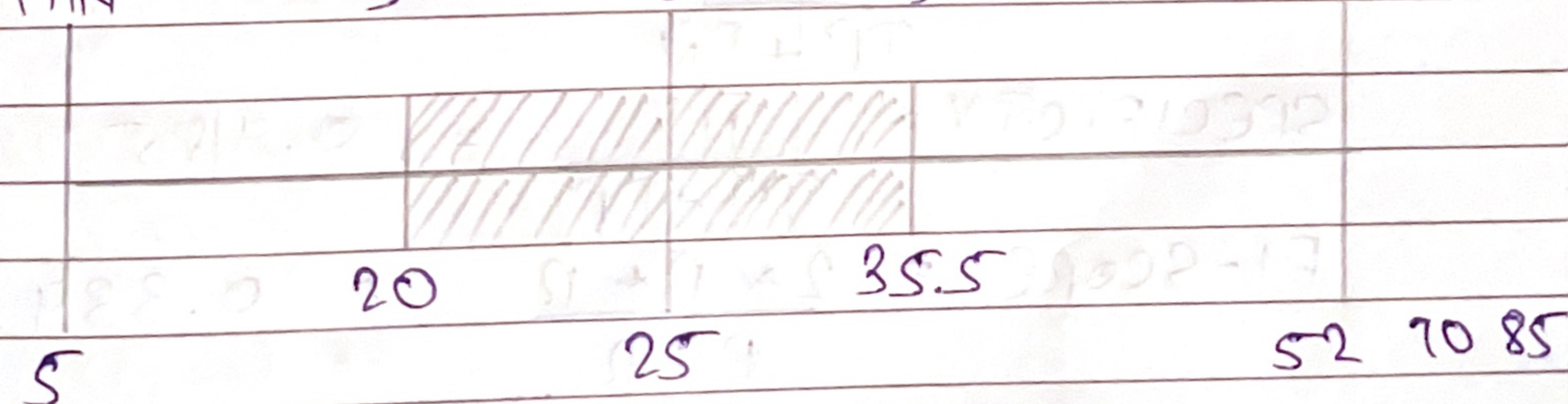
(IQR) = $Q3 - Q1 = 35.5 - 20 = 15.5$

UPPER BOUND = $Q3 + 1.5 * IQR = 58.75$

LOWER BOUND = $Q1 - 1.5 * IQR = -3.25$

∴ 70, 85 ARE OUTLIERS

MIN Q1 MEDIAN Q3 MAX



[CONFUSION MATRIX]

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	P(Y)	P(N)
Q. A (Y)	50 TP	10 FN
A (N)	5 FP	35 TN

$$\text{ACCURACY} = \frac{TP + TN}{TP + FP + FN + TN} = \frac{50 + 35}{50 + 5 + 10 + 35} = 0.85$$

$$\text{PRECISION} = \frac{TP}{TP + FP} = \frac{50}{50 + 5} = 0.909$$

$$\text{RECALL} = \frac{TP}{TP + FN} = \frac{50}{50 + 10} = 0.833$$

$$\text{SPECIFICITY} = \frac{TN}{TN + FP} = \frac{35}{35 + 5} = 0.875$$

$$\text{F1-SCORE} = \frac{2 * P * R}{P + R} = 0.869$$

	P(Y)	P(N)	T
Q. A (Y)	90 TP	210 FN	300
A (N)	140 FP	9560 TN	9700
T	230	9770	10000

$$\text{ACCURACY} = \frac{TP + TN}{TP + FP + FN + TN} = \frac{90 + 9560}{90 + 140 + 210 + 9560} = 0.965$$

$$\text{PRECISION} = \frac{TP}{TP + FP} = \frac{90}{90 + 140} = 0.391$$

$$\text{RECALL} = \frac{TP}{TP + FN} = \frac{90}{90 + 210} = 0.3$$

$$\text{SPECIFICITY} = \frac{TN}{TN + FP} = \frac{9560}{9560 + 140} = 0.985$$

$$\text{F1-SCORE} = \frac{2 * P * R}{P + R} = 0.339$$

[APRIORI]

Q.	TRANSACTION ID	ITEM BOUGHT
	1	M, B
	2	M, D
	3	M, B, D
	4	B, D
	5	M, B

GIVEN $n = 5$
min. supp. = $5 * 60\% = 3$

①	ITEM	COUNT	REMARK
	M	4	ACCEPTED
	B	3	ACCEPTED
	D	3	ACCEPTED

MAKING PAIRS OF ITEMS:

②	ITEMS	COUNT	REMARK
	M, B	3	ACCEPTED
	M, D	2	REJECTED
	B, D	2	REJECTED

GIVEN ONLY TWO ITEMS REMAIN {M, B}
min. conf = $75\% = 0.75$

ASSOCIATION RULES:

$$\textcircled{1} M \Rightarrow B$$

$$\text{conf.} = (M, B / M) \cdot 100 = (3/4) \cdot 100 = 75\%$$

$$\textcircled{2} B \Rightarrow M$$

$$\text{conf.} = (M, B / B) \cdot 100 = (3/4) \cdot 100 = 75\%$$

[APRIORI]

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Q.	TRANSACTION ID	ITEMS
	1	1, 3, 4, 6
	2	2, 3, 5, 7
	3	1, 2, 3, 5, 8
	4	2, 5, 9, 10
	5	1, 4

GIVEN $n = 5$

min. supp. = $5 \times 30\% = 1.5$

①	ITEM	COUNT	REMARK
	1	3	ACCEPTED
	2	3	ACCEPTED
	3	3	ACCEPTED
	4	2	ACCEPTED
	5	3	ACCEPTED
	6	1	REJECTED
	7	1	REJECTED
	8	1	REJECTED
	9	1	REJECTED
	10	1	REJECTED

1. MAKING PAIRS OF ITEMS :

②	ITEM	COUNT	REMARK
	1, 2	1	REJECTED
	1, 3	2	ACCEPTED
	1, 4	2	ACCEPTED
	1, 5	1	REJECTED
	2, 3	2	ACCEPTED
	2, 4	0	REJECTED
	2, 5	3	ACCEPTED

3, 4	1	REJECTED
3, 5	2	ACCEPTED
4, 5	0	REJECTED

MAKING TRIOS OF ITEMS :

ITEMS	COUNT	REMARK
(3) 1, 3, 4	1	REJECTED
1, 2, 3	1	REJECTED
1, 3, 5	1	REJECTED
2, 3, 5	2	ACCEPTED

GIVEN ONLY $\{2, 3, 5\}$ IS ACCEPTED
 min. conf. = 70% = 0.7

ASSOCIATION RULES :

- a) $2, 3 \Rightarrow 5 = (2/2) \cdot 100 = 100\%$
- b) $3, 5 \Rightarrow 2 = (2/2) \cdot 100 = 100\%$
- c) $2, 5 \Rightarrow 3 = (2/3) \cdot 100 = 66.6\%$
- d) $5 \Rightarrow 2, 3 = (2/2) \cdot 100 = 100\%$
- e) $2 \Rightarrow 3, 5 = (2/3) \cdot 100 = 66.6\%$
- f) $3 \Rightarrow 2, 5 = (2/3) \cdot 100 = 66.6\%$

$\Rightarrow$ THE ASSOCIATION RULES $(2, 3) \Rightarrow 5$, $(3, 5) \Rightarrow 2$ ARE STRONG IF MINIMUM CONFIDENCE THRESHOLD IS 70% AS BOTH ARE 100% THAT IS GREATER THAN OR EQUAL TO 70% THRESHOLD

[NAME BAYES]

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Q. IS IT ADVISABLE TO PLAY ON A DAY WITH HOT TEMP., RAINY OUTLOOK, HIGH HUMIDITY AND NO WIND?

Let,

SUNNY $\Rightarrow$ S

OVERCAST $\Rightarrow$ O

RAIN $\Rightarrow$ R

HOT $\Rightarrow$ H

MILD $\Rightarrow$ M

COOL $\Rightarrow$ C

HIGH $\Rightarrow$ H

NORMAL $\Rightarrow$ N

FALSE $\Rightarrow$ F

TRUE $\Rightarrow$ T

NO $\Rightarrow$ N

PLAY $\Rightarrow$ Y

$$n(\text{No}) = 5$$

$$n(\text{Yes}) = 9$$

$$n = 14$$

ATTRIBUTES :

- TEMP.

- OUTLOOK

- HUMIDITY

- WINDY

- CLASS

OUTLOOK	TEMP.	HUMID.	WIND	CLASS
S	H	H	F	N
S	H	H	T	N
O	H	H	F	Y
R	M	H	F	Y
R	C	N	F	Y
R	C	N	T	N
O	C	N	T	Y
S	M	H	F	N
S	C	N	F	Y
R	M	N	F	Y
S	M	N	T	Y
O	M	H	T	Y
O	H	N	F	Y
R	M	H	T	N

Let,

$$C1 = \text{CLASS } (Y) \therefore P(Y) = 9/14$$

$$C2 = \text{CLASS } (N) \therefore P(N) = 5/14$$

$$x_1 \Rightarrow P(\text{outlook } (R), n(Y)) = 3/9$$

$$x_2 \Rightarrow P(\text{temp. } (H), n(Y)) = 2/9$$

$$x_3 \Rightarrow P(\text{humid } (H), n(Y)) = 3/9$$

$$x_4 \Rightarrow P(\text{wind } (F), n(Y)) = 6/9$$

$$P(X, n(Y)) = x_1 \cdot x_2 \cdot x_3 \cdot x_4$$

$$= \frac{3}{9} \cdot \frac{2}{9} \cdot \frac{3}{9} \cdot \frac{6}{9} = \frac{4}{243}$$

$$\alpha \Rightarrow P(X, n(Y)) \cdot P(Y) = \frac{4}{243} \cdot \frac{9}{14} = 0.01058$$

$$y_1 \Rightarrow P(\text{outlook } (R), n(N)) = 2/5$$

$$y_2 \Rightarrow P(\text{temp. } (H), n(N)) = 2/5$$

$$y_3 \Rightarrow P(\text{humid } (H), n(N)) = 4/5$$

$$y_4 \Rightarrow P(\text{wind } (F), n(N)) = 2/5$$

$$P(Y, n(N)) = y_1 \cdot y_2 \cdot y_3 \cdot y_4$$

$$= \frac{2}{5} \cdot \frac{2}{5} \cdot \frac{4}{5} \cdot \frac{2}{5} = \frac{32}{625}$$

$$\beta \Rightarrow P(Y, n(N)) \cdot P(N) = \frac{32}{625} \cdot \frac{5}{14} = 0.01828$$

COMPARING α & β ,

$$\alpha < \beta$$

$\therefore$ IT IS NOT ADVISABLE PLAY TENNIS
FOR HOT TEMP., RAINY OUTLOOK,
HIGH HUMIDITY AND NO WIND

[NORMALIZATION]

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NOTE: ALWAYS SORT THE DATA

Q. DATA = [10, 20, 30, 40, 50]

= MIN-MAX NORMALIZATION [0, 1] RANGE:
HERE,

$$x_{\min} = 10, \quad x_{\max} = 50$$

$$X = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad \text{--- FORMULA}$$

x	NORMALIZED (X)
10	0
20	0.25
30	0.5
40	0.75
50	1

= Z-SCORE NORMALIZATION:
HERE,

$$\mu = 30, \quad \sigma = 14.14$$

$$Z = \frac{x - \mu}{\sigma} \quad \text{--- FORMULA}$$

x	NORMALIZED (Z)
10	-1.414
20	-0.707
30	0
40	0.707
50	1.414

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{n}} \quad \text{--- STANDARD DEVIATION}$$

[K-MEANS - 2 DIMENSION]

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HERE,

$$n = 8$$

$$k = 3$$

$$C1 = (2, 10), C2 = (5, 8), C3 = (1, 2)$$

WE KNOW THAT

$$D = |x_2 - x_1| + |y_2 - y_1|$$

(1)

	x, y	$C1$	$C2$	$C3$	Closest
A1	2, 10	0	5	9	$C1$
A2	2, 5	5	6	4	$C3$
A3	8, 4	12	7	9	$C2$
A4	5, 8	5	0	10	$C2$
A5	7, 5	10	5	9	$C2$
A6	6, 4	10	5	7	$C2$
A7	1, 2	9	10	0	$C3$
A8	4, 9	3	2	10	$C2$

CLUSTER 1 $\Rightarrow \{A1\}$

CLUSTER 2 $\Rightarrow \{A3, A4, A5, A6, A8\}$

CLUSTER 3 $\Rightarrow \{A2, A7\}$

= CALCULATING NEW CENTROIDS:

$$C1(x) = 2$$

$$C1(y) = 10$$

$$C2(x) = (8 + 5 + 7 + 6 + 4) / 5 = 6$$

$$C2(y) = (4 + 8 + 5 + 4 + 9) / 5 = 6$$

$$C3(x) = (2 + 1) / 2 = 1.5$$

$$C3(y) = (5 + 2) / 2 = 3.5$$

NOW,

$$C1 = (2, 10)$$

$$C2 = (6, 6)$$

$$C3 = (1.5, 3.5)$$

(2)

	x, y	$C1$	$C2$	$C3$	Closest
A1	2, 10	0	8	7	C1
A2	2, 5	5	5	2	C3
A3	8, 4	12	4	7	C2
A4	5, 8	5	3	8	C2
A5	7, 5	10	2	7	C2
A6	6, 4	10	2	5	C2
A7	1, 2	9	9	2	C3
A8	4, 9	3	5	8	C1

CLUSTER 1 $\Rightarrow \{A1, A8\}$ CLUSTER 2 $\Rightarrow \{A3, A4, A5, A6\}$ CLUSTER 3 $\Rightarrow \{A2, A7\}$

=

CALCULATING NEW CENTROIDS:

$$C1(x) = (2+4)/2 = 3$$

$$C1(y) = (10+9)/2 = 9.5$$

$$C2(x) = (8+5+7+6)/4 = 6.5$$

$$C2(y) = (4+8+5+4)/4 = 5.25$$

$$C3(x) = (2+1)/2 = 1.5$$

$$C3(y) = (5+2)/2 = 3.5$$

 $\Rightarrow$

$$C1 = (3, 9.5), C2 = (6.5, 5.25), C3 = (1.5, 3.5)$$

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Q.

OBJECT	x	y
A	1	1
B	1.5	1.5
C	5	5
D	3	4
E	4	4
F	3	3.5

FINDING DISTANCE BETWEEN EACH

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

①

	A	B	C	D	E	F
A	0	0.70	5.65	3.60	4.24	3.20
B		0	4.94	2.91	3.53	2.5
C			0	2.23	1.41	2.5
D				0	1	0.5
E					0	1.1
F						0

= "F" MERGES INTO "D"

②

	A	B	C	DF	E
A	0	0.70	5.65	3.60	4.24
B		0	4.94	2.91	3.53
C			0	2.23	1.41
DF				0	1
E					0

= "B" MERGES INTO "A"

③

	AB	C	DF	E
AB	0	5.65	3.60	4.24
C		0	2.23	3.53 1.14
DF			0	1
E				0

= "E" MERGES INTO "DF"

④

	AB	C	DEF
AB	0	5.65	3.60
C		0	2.23
DEF			0

= "DEF" MERGES INTO "C"

⑤

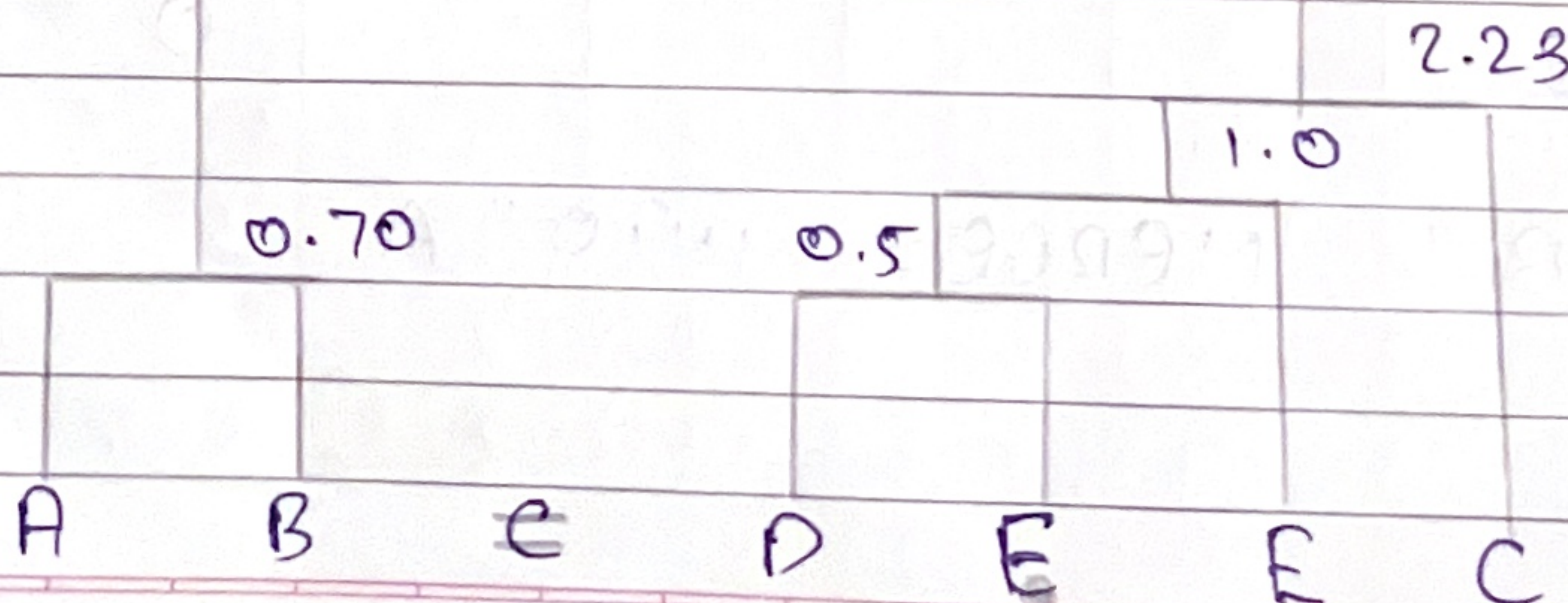
	AB	CDEF
AB	0	5.65
CDEF		0

= "CDEF" MERGES INTO "AB"

⇒ (D, F) → (A, B) → ((D, F), E)
 → (((D, F), E), C) → (((A, B), (D, F), E), C)

5.65

[DENDOGRAM]



[BINNING]

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DATA = [45, 50, 55, 58, 60,
65, 70, 72, 75, 80]

HERE,

MIN = 45

MAX = 80

$\therefore$ RANGE = MAX - MIN = 80 - 45 = 35

BIN RANGES: (BIN WIDTH = $35/3$)

BIN 1 $\Rightarrow$ 45 - 56.667

BIN 2 $\Rightarrow$ 56.66 - 68.333

BIN 3 $\Rightarrow$ 68.333 - 80

=

FREQUENCY

COUNTS:

BIN

ITEMS

COUNT

BIN 1

45, 50, 55

3

BIN 2

58, 60, 65

3

BIN 3

70, 72, 75, 80

4

=

(HISTOGRAM) GRAPH

